

Visual ERP-based brain-computer interface use with severe physical, speech and eye movement impairments: case studies

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Abstract

Background: Individuals with severe speech and physical impairment face significant challenges in communication and daily interaction. Visual brain-computer interfaces (BCIs) offer a potential assistive solution, but their usability is limited when facing eye motor impairments that affect gaze redirection and fixation. This study investigates the feasibility of a gaze-independent visual oddball BCI for individuals with severe eye motor impairment for use with augmentative and alternative communication (AAC). *Methods:* Seven participants with varying degrees of eye motor impairments were recruited and visual oddball BCI decoding accuracy was evaluated under three conditions: overt, covert, and free visuospatial attention (VSA) using multiple classification methods. *Results:* covert VSA with central fixation leads to decreased accuracy, whereas free VSA is comparable to overt VSA in some gaze-impaired participants. Furthermore, cross-condition decoder training and evaluation suggests that training with overt VSA may improve performance in gaze-impaired individuals. *Conclusions:* These findings highlight the need for adaptive decoding strategies

and further validation in applied settings in the presence of gaze impairment.

Keywords: brain-computer interface, augmentative and alternative communication (AAC), event-related potentials, covert visuospatial attention, eye motor impairment

1 Background

Neurological conditions, such as acquired brain lesions, neuromuscular disorders and Amyotrophic Lateral Sclerosis (ALS) can result in severe speech and physical impairment. This, in turn, significantly alters an individual’s ability to communicate and interact with their environment, reducing the level of activity and participation, and overall quality of life. Augmentative and alternative communication (AAC) technology [1, 2, 3] leveraging visual brain-computer interfaces (BCIs)[4, 5], which relies on the interpretation of visual stimuli by the user, offers several advantages in this context. The rapid stimulation pace of visual BCIs and the modulation of information over spatial attention allow for high information transfer rates (ITRs) [6, 7]. Combined with their ability to operate with non-invasive recording technology, this makes them well-suited for communication tasks.

However, severe visual impairments such as nystagmus (uncontrolled eye movements), diplopia (double vision), ophthalmoplegia (paralysis), fatigability and head motion limitations can significantly hamper an individual’s ability to accurately control eye gaze, reducing their performance with BCIs that rely on visual cues [8, 9, 10, 11]. Unfortunately, it is again for this group that eye tracking solutions also perform poorly, making them more reliant on potential developments in BCIs that do not rely on eye gaze.

Hence, some BCI and AAC solutions rely on the capability of the user to direct their gaze at a desired target. In this case, the gaze fixation and the visuospatial attention (VSA) of the user coincide. Alternative solutions usually circumvented by designing gaze-independent BCIs [12], which either avoid visual stimulation or exploit some form of covert VSA, where gaze and VSA do not coincide.

Several studies on visual oddball BCIs indicate that performance declines when relying on covert VSA [13, 14, 15]. While these studies were performed with healthy participants, their ultimate aim was to discover obstacles preventing accurate, gaze-independent decoding for individuals with gaze impairment. To that end, they emulate covert VSA using central gaze fixation. This implicitly assumes that BCI users with eye motor impairment who would benefit from gaze-independence would feel comfortable performing this central fixation. In other words, one could argue that a BCI verified to work only with central fixation could also be considered gaze-dependent. It does not account for the residual eye motor capabilities of most individuals with severe speech, physical and gaze impairment (SSPGI), the (dis)comfort they experience

while performing gaze fixation, and other confounding factors resulting from their eye motility.

It is notable that studies reporting on gaze-independent visual BCI use by individuals with SSPGI are scarce. Results are usually different from those obtained with healthy control participants, due to difference in capabilities, brain response, equipment and environment.

Combaz et al. [16] showed that steady-state visually evoked potential (SSVEP) and oddball paradigm BCIs can be used effectively by individuals with Locked-in Syndrome (LiS). Lesenfants et al. [17] tested a BCI using gaze-independent SSVEPs in six participants with LiS yet only two of them exceeded chance level accuracy. More recently, Peters et al. [18] performed a trial with two participants at a later stage of ALS with severe visual impairment. Their SSVEP paradigm was not optimized for gaze-independence, but the system showed high accuracy, outperforming an eye tracking alternative. It would be of interest to determine whether such results can be replicated with participants with other neurological conditions, and with a visual oddball rather than an SSVEP paradigm.

Orhan et al. [19] and Oken et al. [20] tested the rapid serial visual presentation (RSVP) speller with individuals with LiS. Due to the serial nature of the stimulation paradigm, communication was rather slow but they demonstrated viability of the RSVP paradigm in a relevant target population.

Severens et al. [21] evaluated the visual Hex-o-Spell [14] on 5 participants with ALS and showed that this visual oddball interface optimized for gaze-independence can outperform a tactile BCI. While this speaks to the power of visual paradigms even in groups that are expected to have eye motor impairment, they did not verify the gaze direction of participants during the experiment. It was assumed that participants were performing overtly. Participants with ALS also exhibited a substantially lower accuracy than healthy controls (58% vs. 88%).

Van Den Kerchove et al. [22] also built towards a gaze-independent solution using the visual Hex-o-Spell interface. They partially accounted for the fact that BCI users with SSPGI might not fully rely on central gaze fixation by evaluating settings independent of central fixation. They showed gaze-independent performance can be improved in healthy subjects by using a suited decoding strategy that corrects for latency jitter in covert VSA responses. Yet, there is a need for verification of these results in individuals with SSPGI.

Ultimately, this research aims to develop gaze-independent BCI for individuals who are fully locked-in and have no alternative means of communication. However, this group is very small and it is often challenging to recruit them for a study and they may experience challenges in performing the experiments [23]. Individuals with less severe paralysis or in less progressed disease stages that struggle with eye-tracking technology could also benefit from solutions tailored to their specific situation and residual capabilities. Therefore, we apply the concepts from earlier work and literature to individuals with severe speech and physical impairment affected by various degrees of eye motor impairment using a visual oddball BCI. The contributions of this

study are as follows: (1) the recruitment of individuals that are specifically affected with SSPGI in a visual BCI study, (2) the exploration of their eye motility and experienced comfort when operating such a BCI, (3) the evaluation of the performance of a gaze-independent visual BCI for this group, and (4) the verification of gaze-independent decoding strategy proposed by Van Den Kerchove et al. [22]. Finally, we formulate a set of recommendations based on our experience when carrying out this study, which could be used in designing further similar studies or BCI-AAC solutions for this target user population.

2 Methods

2.1 Recruitment

Participants were recruited from the Neuromuscular Reference center at University Hospitals Leuven (Leuven, Belgium), TRAINM Clinics (Antwerp, Belgium), the Neurorehabilitation Unit at the Lille University Medical Center (Lille, France), and Fondation Partage et Vie (Loos, France). Experiments were performed after prior informed consent and under the supervision of the treating physician, with a protocol prior approved by the Ethics Commission of the University Hospitals Leuven (S62547).

We partially base the recruitment criteria on the BCI user classification system presented by Wolpaw et al. [23]. They present a three-class system: (1) complete LiS without “useful neuromuscular control,” including eye movements, (2) the ability to control a single-button switch using very limited neuromuscular control consisting of eye movements or limited limb movements, (3) impaired but substantial neuromuscular control including speech or hand control, able to operate a range of classical AAC devices.

In order to be included in the study, participants must:

1. be at least 18 years old and no older than 60 years,
2. belong to class 2 or 3 according to the aforementioned BCI user classes,
3. have limitations to the extent or comfort of their eye motor control (partial or full gaze paralysis, uncontrolled gaze movements, or conditions affecting the capability to direct the gaze or fixate),
4. have given their informed consent prior to participation.

Participants were excluded if they:

1. had any diagnosis of a major medical condition other than neuromuscular conditions covered by inclusion criteria 2 and 3,
2. had a predisposition to or a history of any kind of epileptic seizures, including photosensitive epilepsy,

ID	Diagnosis	Age	Sex	Speech	Primary communication method	W	KB
PA1	bulbar-onset ALS	58	M	anarthric	tablet with touchscreen	3	4
PB1	FRDA	41	M	dysarthric	speech	3	3
PB2	FRDA	43	F	dysarthric	speech	3	3
PB4	FRDA	48	M	dysarthric	speech	3	3
PC2	ischemic brainstem stroke	43	M	anarthric	eye movements	2	4
PC3	hemorrhagic brainstem stroke	43	F	anarthric	letterboard with finger pointing or touch	2	3
PC4	ischemic brainstem stroke	54	M	anarthric	letterboard with finger pointing or touch	2	3

Table 1: Included participants with their diagnosis and relevant communication capabilities. W: classification according to Wolpaw et al. [23] used as inclusion criterion, KB: classification according to Kübler and Birbaumer [24] for further reference.

3. had a severe loss in vision or hearing that would significantly impair participation in the experiment,
4. were using specific psychoactive medications or substances that could affect the outcome (neuroleptics or benzodiazepines),
5. were unable to understand the experiment instructions or to cooperate,
6. had any other limitations preventing them from performing the given task.

In total, 11 individuals were contacted. Of these, one person with Multiple Sclerosis (MS) was excluded based on criterion 3. A person recovering from traumatic brain injury (TBI) was excluded based on criteria 2 and 4, and one person recovering from stroke based on criterion 1. One further person recovering from a stroke was excluded due to technical difficulties during the experimental session.

Ultimately, 7 participants were retained. Of these, one participant was diagnosed with bulbar-onset ALS, three with Friedreich’s Ataxia (FRDA) and three were recovering from LiS due to stroke. Table 1 lists the included participants and their diagnoses, each case is clinically described in appendix A.

2.2 Visual skill and eye motor examination

Self-reported eye motor and visual abnormalities or those observed in clinical assessment are reported in table 2. Visual skills were assessed using the framework established by Fried-Oken et al. [9]. These describe on a higher level the factors that can reduce visual BCI performance when impaired, and include visual acuity, visual fixation, eyelid function, ocular motility, binocular vision, and field of vision. Additionally, participants and their caregivers were asked about eye tremors (nystagmus or other) and other involuntary eye movements. Vision was assessed using a logMAR chart [25]. During the assessment of visual skills and during the rest of the experimentation session, participants wore their corrective glasses or lenses where applicable.

All participants reported some degree of fatigue or discomfort when fixating. Participant PA1 had the mildest impairment, only reporting fatigue when fixating for prolonged times. The FRDA participants were

	PA1	PB1	PB2	PB4	PC2	PC3	PC4
Visual fixation	—	—	—	—	—	—	—
Eyelid function	+	+	+	+	+	—	—
Ocular motility	+	—	+	—	/	/	—
Binocular vision	+	+	+	+	—	/	/
Field of vision	+	+	+	+	+	/	—
Involuntary movement	+	—	/	—	—	—	+
Visual acuity	0.0	0.0	0.6	0.2	0.0	0.0	0.6

Table 2: Self-reported visual skills as defined by Fried-Oken et al. [9] of gaze-impaired participants included in this study. + skilled, — impaired, / severely impaired. Visual acuity was assessed using the logMAR scale (0.0: normal vision, 0.05-0.5: moderate visualimpairment, 0.5-1.3: severe visual impairment, > 1.3 blindness [26]).

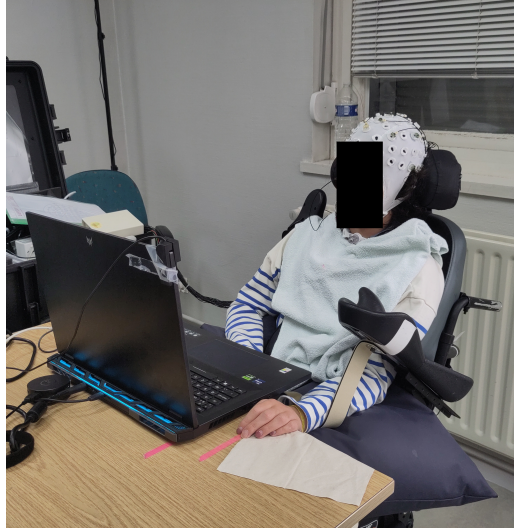
mostly affected by eye tremors and impaired pursuit. Eye motor function of participants PC2, PC3, and PC4, those recovering from brain stem and cerebellar stroke, was most severely affected.

Finally, we also recorded gaze position throughout the experimental session to register the participant’s gaze relative to the stimulated BCI targets, using a Tobii X2-30 Compact (Tobii Technology AB, Sweden) portable eye tracker placed at the bottom of the laptop screen. The registered data was cleaned by fusing left and right eye screen-based gaze coordinates into one channel for the horizontal and one for vertical gaze position. If both were present for a given sample, the fused channel was the mean of both values. If at a given sample either the left or the right eye was not detected for a given channel, the value of the other one was adopted. If both were missing, the gaze position remained unset at that time point, and no interpolation was performed.

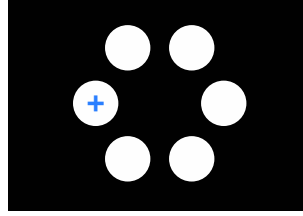
2.3 BCI stimulation

The BCI stimulation procedure was based on the Hex-o-Spell [14] implementation presented by Van Den Kerchove et al. [22]. Similar to this study, the task consists of counting the intensification of a cued target among 6 circular targets laid out in a hexagonal pattern in the field of view of the user as displayed in fig. 1. We refer to Van Den Kerchove et al. [22] for further details on the implementation of the stimulation procedure.

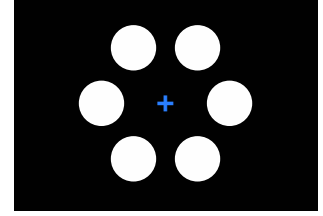
Three different VSA settings were explored. In the overt VSA setting, the participant was instructed to fixate on the cued target or to try this to the maximum extent of their visual skill. In the covert VSA setting, the participant was instructed to fixate on the center of the screen, to the extent of their ability. An additional *free VSA* setting was introduced. Here, the participant was instructed to perform the task as they deemed most comfortable. This allowed us to investigate the user’s natural way of operating the BCI given their individual set of visual skills. If the participant was not fully paralyzed, they were instructed not to move their head. The cued *split attention* setting proposed by Van Den Kerchove et al. [22] was not



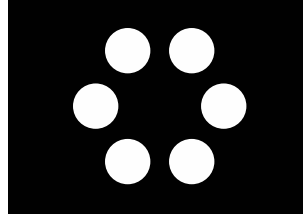
(a) A participant seated in a wheelchair in front of the stimulation laptop with EEG cap.



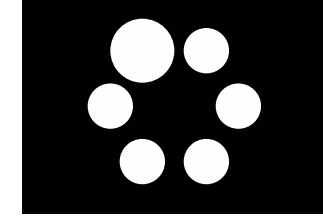
(b) Stimulation interface with 6 targets and fixation crosshair positioned for *overt* VSA.



(c) In *covert* VSA, the fixation crosshair is placed in the center of the screen.



(d) In *free* VSA, no fixation crosshair is displayed.



(e) Targets are intensified by enlarging them for 100 ms.

Figure 1: Stimulation and recording setup for the oddball BCI experiment with different VSA conditions

studied here, as we were interested in natural VSA operation settings for gaze-impaired individuals.

To make the interface suitable for use by individuals with neurological conditions [9], the number of blocks was reduced to 6 per VSA setting. In order to decrease task difficulty, inter-stimulus interval (ISI) was increased to 200 ms with added random jitter uniformly distributed between -50 ms and 50 ms. The experiment also started with a training block with instructions for each condition, where the participant received oral feedback on their counting accuracy to ensure they understood and were able to perform the task.

2.4 Electroencephalography (EEG) data collection & preprocessing

During the recording session, participants were positioned in their wheelchair in front of a table. Stimuli were presented on an Acer Predator Helios laptop with an 18" screen (Acer, Inc., Taiwan) placed at a 60 cm distance. A Cedrus StimTracker (Cedrus Corp., CA, USA) ensured synchronization of stimuli with the recorded EEG.

EEG was recorded at 1000 Hz using the Neuroscan Neuvo portable amplifier (Compumedics Neuroscan, Australia) connected to a second laptop for registration. The EEG headset used 18 active AgCl electrodes (EASYPAC GmbH, Germany) placed on a cap according to the international 10-20 layout. Using electrolyte gel, electrode impedances were reduced below 10 k Ω . Additionally, the electrooculogram (EOG) was recorded.

The EEG was band-pass filtered between 0.5 and 16 Hz. Bad channels were rejected using the RANSAC

algorithm [27] and visual inspection. Next, the EEG was re-referenced to the average of mastoid electrodes TP9 and TP10, and independent component analysis (ICA) was performed to reject artifactual components based on correlation with the EOG or by visual inspection. Epochs were cut from -0.1 to 0.9 s relative to stimulus onset, and no baseline correction was performed.

2.5 BCI decoding

We evaluated the recorded EEG data using the Classifier-based Latency Estimation with Woody iterations (WCBLE) [22] and block-Toeplitz linear discriminant analysis (tLDA) [28] classifiers, as well as the Riemannian approach XDAWNCov+TS+LDA [29]. For WCBLE, a region of interest from 0 ms to 800 ms relative to stimulus onset was used while the epoch was cropped to -100 ms to 900 ms. For the other decoders, the epoch was cropped between 0 ms and 800 ms, which resulted in maximal accuracy. Single-trial target selection accuracies were obtained using 6-fold cross-validation where folds corresponded to stimulation blocks.

3 Results

3.1 Eye tracking analysis

Given their eye motor impairment as listed in table 2, we aimed to clarify the capabilities of the participants in performing overt VSA and central gaze fixation and to assess the relevance of these settings when gaze is not cued (free VSA) Figure 2 maps gaze position relative to the targets across conditions. These results should be interpreted with care, as the eye tracker partially relies on intact eye motility. The participant’s position relative to the eye tracker might have shifted throughout the experimental session despite our best efforts, due to e.g. a regular need for aspiration of the tracheostoma of some participants.

PA1 had relatively intact gaze control and was able to correctly perform the cued overt and covert settings. When gaze was uncued, he fixated on the cued target. This was also mostly the case for PB1, although eye tracking revealed that he chose not to perform central gaze fixation when cued in at least one of the stimulation blocks. We were unable to record his gaze near the bottom-left stimulus position, either due to eye tracker failure or because the participant was not comfortable fixating on this position. Eye tracker calibration did not succeed for subject PB4, but given the transformation of gaze positions to the stimulus space, he was assumed to be overtly performing the free task.

PB2 was able to perform overt VSA and central fixation to some extent, yet eye tracking shows a larger spread in gaze position compared to PA1 and PB1. In the free VSA condition, however, she preferred to

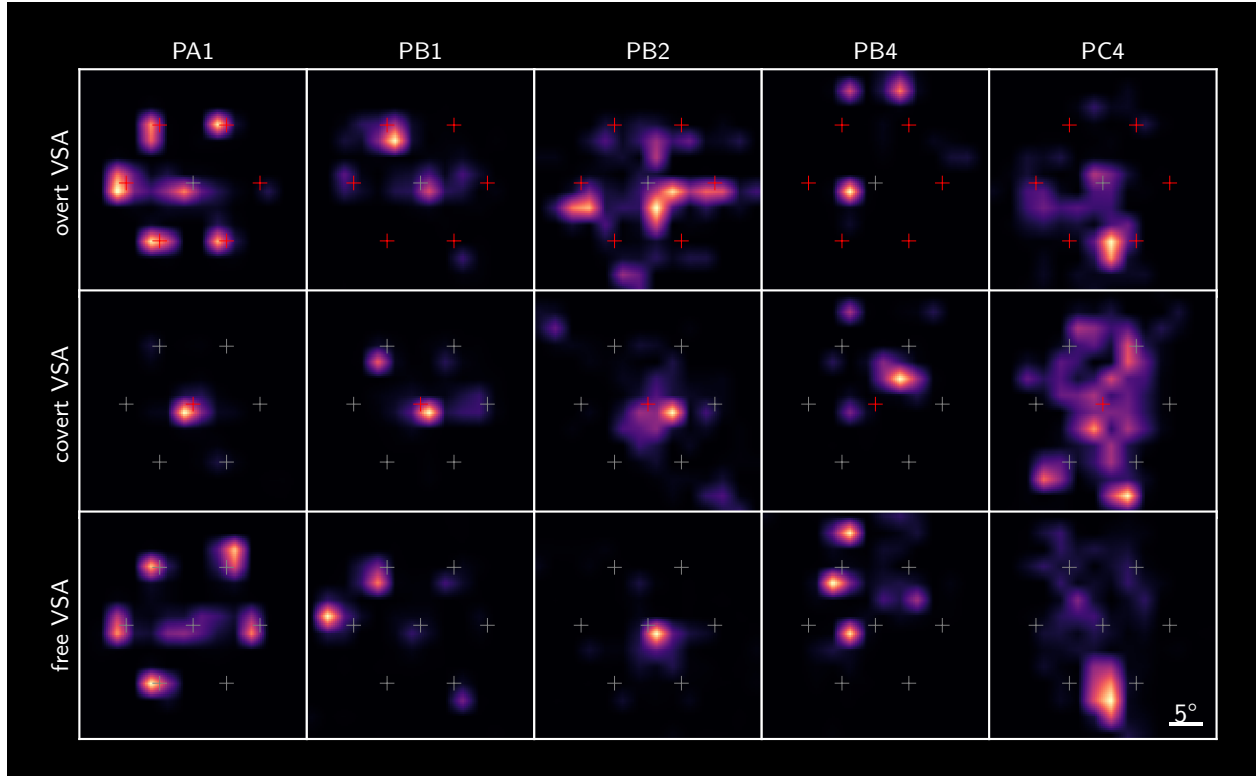


Figure 2: Distribution of the recorded gaze position during the experimental session in the three VSA conditions. Crosshairs represent stimulus positions, with positions cued during the given condition indicated in red. Eye tracking calibration could not be successfully completed for participants PC2 and PC4. Subjects PB2 and PC4 preferred covert BCI operation, with PB2 resting gaze near the middle of the screen, and PC4 near the bottom. A high-resolution, non-downsampled figure is available as Additional file 1.

attend stimuli covertly when the gaze was uncued. This was confirmed by the participant.

The overt and central gaze fixation settings were also not properly adapted to participant PC4. In the free VSA condition, eye tracker results show that his gaze was usually near the bottom two targets, indicating some degree of covert or split VSA.

Technical difficulties were encountered while recording gaze position with the Tobii X2-30 Compact for participants PC2 and PC3, since they both had one eye that was occluded respectively by the prism glass and the eyelid. Both participants reported they could not fixate on some of the stimuli.

3.2 BCI decoding performance

Figure 3 shows cross-validated single-trial target selection accuracy for the evaluated VSA settings for the different decoders. A full table of results is available as Additional file 2. Accuracy results do not reveal a clear trend in decoder performance per condition. In the covert VSA setting with cued central gaze fixation, accuracy deteriorated overall. Target selection accuracy for this task was around chance level ($\frac{1}{6}$) for participants PB4, PC2 and PC3.

WCBLE did not improve over the accuracy of tLDA in the free VSA setting, but XDAWNCov+TS+LDA accuracy was slightly lower here, though not significantly. More interestingly, we noticed that accuracies of the decoders in free VSA were close to those in the overt VSA. A substantial decrease in accuracy from the overt setting to the free setting was observed for subjects PC2, PC3 and PC4 who had the most severe eye motor impairment. For PB2, who relied on covert VSA during the uncued free VSA according to gaze tracking setting, the decrease in accuracy was also present.

3.3 Cross-condition decoder training and evaluation

As an alternative approach to selecting the most suitable decoder, we used tLDA as the base decoder and verified whether accuracy could be improved if BCI users with gaze impairment performed the decoder training session relying maximally on their residual gaze control. A full table of results is available as Additional file 3.

Figure 4 shows that for participants PB1 and PC2 covert VSA decoding improved when training with overt VSA. Note that, according to eye tracking data, participant PB1, PB4, and PC4 did not always perform cued central gaze fixation in the covert VSA setting, which might have affected the results.

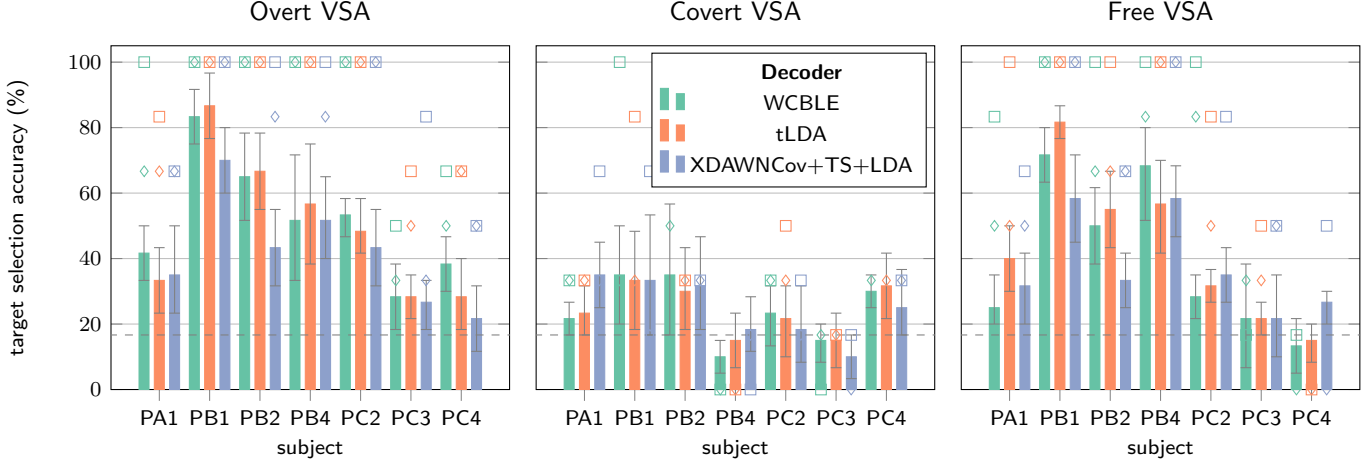


Figure 3: Decoding performance in different VSA settings reported as single-trial target selection accuracy (%). Accuracy in free VSA is generally on par with the overt VSA setting. In the covert VSA setting with central gaze fixation, accuracy is lower, but can be improved with the WCBLE decoder. Whiskers indicate 95% confidence intervals determined using 10,000 bootstrapping repetitions. Dashed line indicates the chance level accuracy. Diamond markers indicate target selection accuracy using the median score for five stimulation repetitions. Square markers indicate target selection accuracy for 10 stimulation repetitions. ($\frac{1}{6}$).

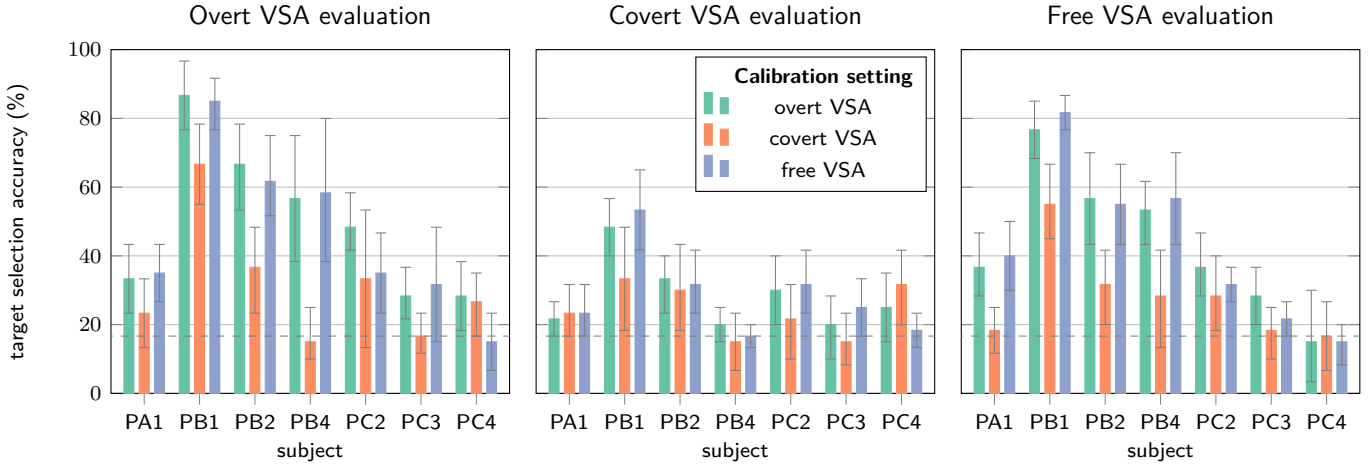


Figure 4: Decoding performance when calibrating the tLDA decoder in a given VSA setting, and evaluating it in another, reported as single-trial area under the receiver-operator characteristic curve. Whiskers indicate 95% confidence intervals determined using 10,000 bootstrapping repetitions. For participants PA1, PB2, and PC3, decoding accuracy in the covert VSA setting with central gaze fixation improved when calibrating with overt gaze fixation.

4 Discussion

This study investigated the feasibility of a gaze-independent visual oddball BCIs for individuals with SSPGI. While covert VSA with cued, central gaze fixation resulted in reduced decoding accuracy, likely due to an increased task load, accuracy in free VSA where gaze position is uncued can yield performance comparable to explicitly cued overt VSA for some participants. This suggests that the evaluated participants naturally integrate residual gaze control. Decoder training with cued, overt VSA can improve accuracy in subsequent free VSA conditions, highlighting the potential benefits of leveraging this residual eye motor control during the decoder training phase. While gaze-independent BCIs show promise for individuals with SSPGI, usability and comfort remain critical considerations for real-world applications. These results provide important insights into the optimization of visual BCIs for individuals with severe motor impairments and underscore the need for further research into adaptive decoding strategies and user-centered design.

4.1 Gaze-independent operation & decoding

Due to the heterogeneous nature of the participants' conditions and the limited number of participants, it is difficult to draw general conclusions. This study should therefore be seen as a collection of case studies, highlighting different obstacles encountered in developing gaze-independent visual oddball BCIs for individuals with SSPGI. Nevertheless, we highlight some aspects that might be of interest for the further development of this class of BCIs.

True *gaze-independent* visual BCIs should not rely on gaze fixation. Hence, our analysis centers around the free VSA condition. Eye tracking results presented in section 3.1 confirm our assumption that voluntary covert VSA can occur in individuals with SSPGI. We also confirmed part of the results presented by Van Den Kerchove et al. [22], which state that decoding of covert VSA with central gaze fixation can be improved by accounting for latency jitter. We showed that this also holds for some individuals with SSPGI, with PA1 and PC4 as examples.

Contrary to our assumptions, we have shown that accounting for latency jitter does not necessarily improve covert VSA when gaze fixation is not cued. One possible explanation is that actively performing central gaze fixation increases task load. This, in turn, can reduce overall performance, even though the participant might have otherwise performed covert VSA, but would not be occupied with maintaining strict central gaze fixation. This extra task demand is not present in the free VSA condition, so there is less performance to be gained. Furthermore, cued central gaze fixation combined with counting flashing stimuli in the visual periphery is an explicit example of a dual task. Dual tasks have been shown to increase latency jitter of the P3 event-related potential (ERP) component [30, 31, 22], which is what WCBLE accounts for.

Hence, increased P3 jitter might be more related to maintaining central gaze fixation than to the actual covert VSA aspect.

The seemingly stable accuracy across overt and free VSA could be misinterpreted as an indication that the Hex-o-Spell BCI already works well for individuals with SSPGI, and no optimization is needed. However, we assume that overt VSA accuracy was also decreased in some subjects or for some blocks if the participant was not able to comfortably perform the task. Nevertheless, the large difference between the free VSA setting and the covert VSA setting with central gaze fixation is food for thought about the applicability of solutions developed with central fixation in mind.

Individuals with all except the most severe gaze impairments will likely retain some capability to direct their gaze in visual BCI operation, which can drastically boost accuracy. Subject PB2 exemplifies this: his free VSA decoding accuracy is above that in covert VSA, and eye tracking showed that he relied mostly on overt VSA when cued to do so, and mostly on covert VSA when gaze was uncued. This is also supported by our results on cross-condition decoder training and evaluation presented in section 3.3, which show that leveraging residual eye motor control to fixate targets during the decoder training phase can improve decoding accuracy in some settings. This is likely due to the increased P3 component amplitude in overt VSA, which improves the discriminative power of a classifier trained on this data. Cueing this overt gaze fixation only during the decoder training phase leaves the user free to operate in the manner that is most comfortable in the operation phase. The presence of early, visual ERP components in the training data can then also contribute to performance in those cases where the participant was not able to perform covert VSA with central gaze fixation.

4.2 Clinical implications

The population of individuals with SSPGI is sparse, yet is regularly confronted with major challenges. As opposed to individuals in a vegetative or severe minimally conscious state, they demonstrably have the intent and capability to communicate their thoughts and desires to their clinicians, caregivers and social network. These capabilities, however, are severely limited by their condition, reducing the effectiveness and efficiency of communication. Hence, finding a way to fill in this gap is a major issue in the care of individuals with SSPGI.

Our work shows that some AAC users might benefit from visual BCIs for home use or in the clinical setting. While the proposed communication protocol is a proof of concept with a limited degree of freedom, it is a step towards applications like textual communication and environment or home automation control that inherits the relatively high information transfer rate of visual BCIs.

Furthermore, our experiments revealed that the required technology and its potential applications were generally well received by the participants and their environment. While the necessary visual attention task can be taxing if performed for extended periods of time, participants indicated that this was outweighed by the potential to communicate in a more automated and autonomous way compared to their current AAC solutions, which often required the help of a trained caregiver.

4.3 Limitations & recommendations

Despite the insights gained on gaze-independent BCI approaches, certain limitations should be noted. First and foremost, this study works with a limited sample size, which does not represent the full spectrum of individuals with SSPGI and their specific symptoms and residual skills. Individuals with FRDA met the inclusion criteria, but they are usually not considered a typical interest groups for BCI-AAC, partly due to the rarity of the disease and partly due to the characteristics of its progression and symptoms. It would be most interesting to verify these results with individuals with complete LiS, who retain no eye movement capability at all.

Another limiting factor is the difficulty experienced in correctly interpreting eye tracker results in studies with individuals with gaze impairments. If eye tracking is possible at all, it is not guaranteed that the user is able to successfully perform the calibration procedure. Further experiments should be carried out with a stationary eye tracker with more advanced capabilities, although systems using a head fixator or headrest should be avoided. This is not practical when working with wheelchair-bound individuals who might have undergone a tracheostomy and may suffer from spasticity. Given different more suited gaze tracking hardware, adapted to the participant’s conditions, analyses could be performed with a finer granularity by assessing the gaze condition on a per-epoch basis.

To properly contextualize performance results, they should be coupled with metrics evaluating the full range of the user’s requirements, with measures of usability, comfort and perceived effort, like the NASA Task Load Index [32]. Decoding accuracy might, after all, be traded off for user comfort. The perception of this type of BCI by the user might also be influenced by performing the experiment in an on-line manner, providing immediate feedback after selection and thus closing the loop. In this study, user comfort in the different conditions was not objectively measured. Instead, it was assumed that participants operated most comfortably in the free VSA condition. Even though participants reported that they could comfortably operate the system, this must be confirmed with more quantitative assessments. Eye motor disability could also have been assessed more quantitatively and objectively [9], using, e.g., the Revised Coma Recovery Scale [33] or the NSUCO oculomotor exam [34], in order to relate them to the decoding or satisfaction

334 results.

335 To optimize BCI user experience, research must extend beyond offline classification performance and
336 address the full scope of user interaction during online use. Offline evaluation, while convenient, fails to
337 capture critical factors such as user engagement, learning effects, and true satisfaction. A user-centered
338 design (UCD) framework offers a structured approach to evaluate effectiveness, efficiency, and satisfaction in
339 realistic settings [35, 36, 37, 38]. This includes transitioning from abstract stimuli to meaningful interfaces
340 providing immediate feedback, and evaluating performance metrics in real-time. Satisfaction, a subjective
341 yet vital component, should be assessed through questionnaires after realistic online use [37]. Decoder
342 development, interface design, and paradigm selection should be co-optimized, as improvements in decoding
343 alone may be insufficient if the paradigm does not align with the user’s abilities and preferences. Ultimately,
344 the goal is to design a usable system for individuals with specific impairments, where longitudinal studies
345 with targeted user groups or individuals are required to evaluate system performance comprehensively.

346 Implementing such a user-centered approach requires early and active involvement of individuals with
347 SSPI throughout the research process [35]. Instead of a traditional bottom-up approach – starting with
348 healthy controls and offline analysis – similar future projects should begin directly with online experiments
349 in the target population, identifying challenges and iteratively optimizing the system [9]. This requires long-
350 term collaboration with patient centers to ensure an involved team of clinicians, occupational and speech
351 therapists, and other caregivers, next to further arrangements like ethical approval, and access to suitable
352 infrastructure from the onset. Maintaining a functional in-house online BCI-AAC system facilitates iterative
353 development and realistic performance estimation, ensuring experimental settings reflect practical use. While
354 this top-down approach may not suit novel BCI paradigms in early development, it is appropriate for mature
355 paradigms such as visual oddball BCIs [39, 40], where the main challenge lies in usability and disseminating
356 the technology to potential users.

357 5 Conclusion

358 This study explored the usability of electroencephalography (EEG)-based gaze-independent visual BCIs in
359 individuals with severe speech, physical and gaze impairment (SSPGI), focusing on the impact of eye mo-
360 tor impairments on performance. Our results demonstrate that a visual brain-computer interface (BCI)
361 gaze-independent operation is feasible for some of these individuals. They might either achieve sufficient de-
362 coding accuracy despite their eye motor impairment, or performance can be enhanced by careful, individual
363 adaptation the decoding strategies. The free visuospatial attention (VSA) condition yielded decoding accu-
364 racy comparable to overt VSA in some participants, suggesting that users may naturally integrate residual

gaze control. Additionally, training under overt VSA conditions improved accuracy in subsequent covert or free VSA operation, highlighting the potential benefits of leveraging residual eye motor capabilities during decoder training.

While these findings are promising, future work should include larger participant groups, refine stimulation paradigms, and incorporate real-time, user-centered assessments of comfort and usability to actively develop usable solutions. Ultimately, optimizing gaze-independent BCI designs could enhance communication options for individuals with severe motor and speech impairments, particularly those who struggle with conventional eye-tracking technologies.

Acronyms

AAC augmentative and alternative communication

ALS Amyotrophic Lateral Sclerosis

BCI brain-computer interface

EEG electroencephalography

EOG electrooculogram

ERP event-related potential

FRDA Friedreich's Ataxia

ICA independent component analysis

ISI inter-stimulus interval

ITR information transfer rate

LiS Locked-in Syndrome

MS Multiple Sclerosis

RSVP rapid serial visual presentation

SSPGI severe speech, physical and gaze impairment

SSVEP steady-state visually evoked potential

TBI traumatic brain injury

tLDA block-Toeplitz linear discriminant analysis

UCD user-centered design

VSA visuospatial attention

WCBLE Classifier-based Latency Estimation with Woody iterations

Declarations

Ethics approval and consent to participate

Ethical approval for this multi-center study was obtained from the Ethics Commission of the University Hospitals Leuven (S62547), and prior informed consent to participate was obtained from all participants included in this study.

Consent for publication

Prior informed consent for publication of the reported data was obtained from all participants included in this study, either directly from the participant or through a legal representative.

Availability of data and materials

The datasets generated and/or analysed during the current study are not publicly available due to the policies in agreement with the ethical approval to protect sensitive personal information, but are available from the corresponding author on reasonable request.

Competing interests

The authors declare no competing interests.

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Author’s contributions

AVDK conceptualized the work and performed the analyses presented this study, under supervision of HSM, FC and MMVH. JM assisted in data collection, conceptualization and data analysis. AW and MDM provided paramedical support in participant recruitment, care and communication. DG assisted in participant recruitment at the University Hospitals Leuven NMRC under clinical supervision of PVD. ES supervised participant recruitment at TRAINM. EA provided clinical supervision during conceptualization, experimentation, supervised recruiting at Fondation Partage et Vie and contributed the neuro-ophthalmologic diagnoses. All authors read and approved the final manuscript.

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Additional files

1. Additional file 1

Format: portable network graphics file (.png)

Title: Heatmap of gaze location per VSA condition.

Description: High-resolution heatmap of the distribution of recorded gaze position during the experimental session in the three VSA conditions. Crosshairs represent stimulus positions, with positions cued during the given condition indicated in red.

2. Additional file 2

Format: comma-separated values file (.csv)

Title: Decoding performance per participant. *Description:* Full table of decoding target selection

accuracies per participant, decoder and VSA condition.

3. Additional file 3

Format: comma-separated values file (.csv)

Title: Cross-condition decoding performance per participant. *Description:* Full table of cross-validated, cross-condition decoding target selection accuracies per participant and decoder.

References

1. Ascari REdOS, Pereira R, and Silva L. Mobile interaction for augmentative and alternative communication: a systematic mapping. *Journal on Interactive Systems* 2018 Aug; 9. DOI: 10.5753/jis.2018.704
2. Elshahar Y, Hu S, Bouazza-Marouf K, Kerr D, and Mansor A. Augmentative and alternative communication (AAC) advances: A review of configurations for individuals with a speech disability. *Sensors* 2019; 19:1911. DOI: 10.3390/s19081911
3. Curtis H, Neate T, and Vazquez Gonzalez C. State of the art in AAC: A systematic review and taxonomy. *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*. 2022 :1–22. DOI: 10.1145/3517428.3544810
4. Schultz T, Wand M, Hueber T, Krusienski DJ, Herff C, and Brumberg JS. Biosignal-based spoken communication: A survey. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 2017; 25:2257–71. DOI: 10.1109/taslp.2017.2752365
5. Peters B, Eddy B, Galvin-McLaughlin D, Betz G, Oken B, and Fried-Oken M. A systematic review of research on augmentative and alternative communication brain-computer interface systems for individuals with disabilities. *Frontiers in human neuroscience* 2022 Jul; 16:952380. DOI: 10.3389/fnhum.2022.952380
6. Abiri R, Borhani S, Sellers EW, Jiang Y, and Zhao X. A comprehensive review of EEG-based brain-computer interface paradigms. *en. J Neural Eng.* 2019 Jan; 16:011001. DOI: 10.1088/1741-2552/aaf12e
7. Han J, Xu M, Xiao X, Yi W, Jung TP, and Ming D. A high-speed hybrid brain-computer interface with more than 200 targets. *J. Neural Eng.* 2023 Jan; 20:016025. DOI: 10.1088/1741-2552/acb105
8. McCane LM, Sellers EW, McFarland DJ, Mak JN, Carmack CS, Zeitlin D, Wolpaw JR, and Vaughan TM. Brain-computer interface (BCI) evaluation in people with amyotrophic lateral sclerosis. *Amyotrophic lateral sclerosis and frontotemporal degeneration* 2014 Feb; 15:207–15. DOI: 10.3109/21678421.2013.865750
9. Fried-Oken M, Kinsella M, Peters B, Eddy B, and Wojciechowski B. Human visual skills for brain-computer interface use: a tutorial. *Disability and Rehabilitation: Assistive Technology* 2020 Jun; 15:799–809. DOI: 10.1080/17483107.2020.1754929
10. Pasqualotto E, Matuz T, Federici S, Ruf CA, Bartl M, Olivetti Belardinelli M, Birbaumer N, and Halder S. Usability and workload of access technology for people with severe motor impairment: a comparison of brain-computer interfacing and eye tracking. *Neurorehabilitation and neural repair* 2015 Mar; 29:950–7. DOI: 10.1177/1545968315575611
11. Van Den Kerchove A, Si-Mohammed H, Cabestaing F, and Van Hulle MM. A visual Brain-Computer Interface for gaze-free communication. PhD thesis. KU Leuven, University of Lille, 2024

12. Riccio A, Mattia D, Simione L, Olivetti M, and Cincotti F. Eye-gaze independent EEG-based brain–computer interfaces for communication. en. *J. Neural Eng.* 2012 Jul; 9:045001. DOI: 10.1088/1741-2560/9/4/045001. Available from: <https://doi.org/10.1088/1741-2560/9/4/045001>
13. Brunner P, Joshi S, Briskin S, Wolpaw JR, Bischof H, and Schalk G. Does the ‘P300’ speller depend on eye gaze? en. *J. Neural Eng.* 2010 Sep; 7:056013. DOI: 10.1088/1741-2560/7/5/056013. Available from: <https://dx.doi.org/10.1088/1741-2560/7/5/056013>
14. Treder MS and Blankertz B. (C)overt attention and visual speller design in an ERP-based brain-computer interface. *Behavioral and brain functions* 2010; 6:1–13. DOI: 10.1186/1744-9081-6-28
15. Ron-Angevin R, Garcia L, Fernández-Rodríguez Á, Saracco J, André JM, and Lespinet-Najib V. Impact of speller size on a visual P300 brain-computer interface (BCI) system under two conditions of constraint for eye movement. *Computational Intelligence and Neuroscience* 2019; 2019
16. Combaz A, Chatelle C, Robben A, Vanhoof G, Goeleven A, Thijs V, Van Hulle MM, and Laureys S. A comparison of two spelling brain-computer interfaces based on visual P3 and SSVEP in locked-in syndrome. *PloS one* 2013 Sep; 8. Ed. by Maurits NM:e73691. DOI: 10.1371/journal.pone.0073691
17. Lesenfans D, Habbal D, Lugo Z, Lebeau M, Horki P, Amico E, Pokorny C, Gómez F, Soddu A, Müller-Putz G, et al. An independent SSVEP-based brain–computer interface in locked-in syndrome. *Journal of neural engineering* 2014 May; 11:035002. DOI: 10.1088/1741-2560/11/3/035002
18. Peters B, Bedrick S, Dudy S, Eddy B, Higger M, Kinsella M, McLaughlin D, Memmott T, Oken B, Quivira F, et al. SSVEP BCI and eye tracking use by individuals with late-stage ALS and visual impairments. *Front. Hum. Neurosci.* 2020 Nov; 14:595890. DOI: 10.3389/fnhum.2020.595890
19. Orhan U, Hild KE, Erdogmus D, Roark B, Oken B, and Fried-Oken M. RSVP keyboard: An EEG based typing interface. *2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. IEEE, 2012 Mar :645–8. DOI: 10.1109/icassp.2012.6287966
20. Oken BS, Orhan U, Roark B, Erdogmus D, Fowler A, Mooney A, Peters B, Miller M, and Fried-Oken MB. Brain–computer interface with language model-electroencephalography fusion for locked-in syndrome. *Neurorehabilitation and neural repair* 2014; 28:387–94
21. Severens M, Van der Waal M, Farquhar J, and Desain P. Comparing tactile and visual gaze-independent brain–computer interfaces in patients with amyotrophic lateral sclerosis and healthy users. *Clinical neurophysiology* 2014 Nov; 125:2297–304. DOI: 10.1016/j.clinph.2014.03.005
22. Van Den Kerchove A, Si-Mohammed H, Van Hulle M, and Cabestaing F. Correcting for ERP latency jitter improves gaze-independent BCI decoding. *J. Neural Eng.* 2024 Jul; 21:046013. DOI: 10.1088/1741-2552/ad5ec0
23. Wolpaw JR, Loeb GE, Allison BZ, Donchin E, Nascimento OF do, Heetderks WJ, Nijboer F, Shain WG, and Turner JN. BCI meeting 2005-workshop on signals and recording methods. *IEEE Transactions on neural systems and rehabilitation engineering* 2006 Jun; 14:138–41. DOI: 10.1109/tnsre.2006.875583
24. Kübler A and Birbaumer N. Brain–computer interfaces and communication in paralysis: Extinction of goal directed thinking in completely paralysed patients? *Clinical neurophysiology* 2008 Nov; 119:2658–66. DOI: 10.1016/j.clinph.2008.06.019

25. Bailey IL and Lovie JE. New design principles for visual acuity letter charts. *Optom Vis Sci.* 1976 Nov; 53:740–5. DOI: 10.1097/00006324-197611000-00006
26. World Health Organization. International statistical classification of diseases and related health problems. 10th ed. 2019. Available from: <https://icd.who.int/browse10>
27. Fischler MA and Bolles RC. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Commun. ACM* 1981 Jun; 24:381–95. DOI: 10.1145/358669.358692. Available from: <https://dl.acm.org/doi/10.1145/358669.358692>
28. Sosulski J and Tangermann M. Introducing Block-Toeplitz Covariance Matrices to Remaster Linear Discriminant Analysis for Event-related Potential Brain-computer Interfaces. *arXiv:2202.02001 [cs, q-bio]* 2022 Feb. *arXiv: 2202.02001*. Available from: <http://arxiv.org/abs/2202.02001> [Accessed on: 2022 Mar 10]
29. Cecotti H, Barachant A, King JR, Bornot JS, and Prasad G. Single-trial detection of event-related fields in MEG from the presentation of happy faces: Results of the Biomag 2016 data challenge. *2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE. IEEE, 2017 Jul :4467–70. DOI: 10.1109/embc.2017.8037848
30. Polich J. Updating P300: An integrative theory of P3a and P3b. *en. Clin. Neurophysiol.* 2007 Oct; 118:2128–48. DOI: 10.1016/j.clinph.2007.04.019. Available from: <https://www.sciencedirect.com/science/article/pii/S1388245707001897>
31. Aricò P, Aloise F, Schettini F, Salinari S, Mattia D, and Cincotti F. Influence of P300 latency jitter on event related potential-based brain–computer interface performance. *en. J Neural Eng.* 2014 May; 11:035008. DOI: 10.1088/1741-2560/11/3/035008. [Accessed on: 2023 Mar 23]
32. Hart SG. NASA-task load index (NASA-TLX); 20 years later. *Proceedings of the human factors and ergonomics society annual meeting*. Vol. 50. 9. Sage publications Sage CA: Los Angeles, CA. American Psychological Association (APA), 2006 :904–8. DOI: 10.1037/e577632012-009
33. Giacino JT, Kalmar K, and Whyte J. The JFK Coma Recovery Scale-Revised: measurement characteristics and diagnostic utility. *Archives of physical medicine and rehabilitation* 2004 Jan; 85:2020–9. DOI: <https://doi.org/10.1080/02699050802403557>
34. Maples W. Northeastern State University College of Optometry'Oculomotor norms. *J Behav Optom* 1992; 3:143–50
35. International Organization for Standardization. Ergonomics of Human System Interaction—Part 210: Human-Centered Design for Interactive Systems (formerly known as 13407). *ISO FDIS* 2009; 9241:2009
36. Schreuder M, Riccio A, Riseti M, Dähne S, Ramsay A, Williamson J, Mattia D, and Tangermann M. User-centered design in brain-computer interfaces – A case study. *Artificial intelligence in medicine* 2013 Oct; 59:71–80. DOI: 10.1016/j.artmed.2013.07.005
37. Kübler A, Holz EM, Riccio A, Zickler C, Kaufmann T, Kleih SC, Staiger-Sälzer P, Desideri L, Hoogerwerf EJ, and Mattia D. The user-centered design as novel perspective for evaluating the usability of BCI-controlled applications. *PloS one* 2014 Dec; 9. Ed. by Federici S:e112392. DOI: 10.1371/journal.pone.0112392
38. Han Y, Ziebell P, Riccio A, and Halder S. Two sides of the same coin: adaptation of BCIs to internal states with user-centered design and electrophysiological features. *Brain-Computer Interfaces* 2022 Mar; 9:102–14. DOI: 10.1080/2326263x.2022.2041294

39. Pan J, Chen X, Ban N, He J, Chen J, and Huang H. Advances in P300 brain–computer interface spellers: toward paradigm design and performance evaluation. *Front. Hum. Neurosci.* 2022 Dec; 16:1077717. DOI: 10.3389/fnhum.2022.1077717
40. Fouad IA, Labib FEZM, Mabrouk MS, Sharawy AA, and Sayed AY. Improving the performance of P300 BCI system using different methods. *Network Modeling Analysis in Health Informatics and Bioinformatics* 2020 Sep; 9:64. DOI: 10.1007/s13721-020-00268-1

A Clinical evaluation of participants

PA1 *Diagnosis:* 58 y.o., originally left-handed male diagnosed with bulbar-onset Amyotrophic Lateral Sclerosis (ALS) 2 years prior at age 56. Recently, symptoms started progressing more on the right side and resulted in immobility of the left limbs and partial immobility of the right limbs, but right arm and finger movement were possible, albeit with effort. *Presentation:* PA1 presented lying reclined in a hospital bed, the experimental setup was positioned on a movable folding table. *Communication:* Due to complete anarthria, PA1 communicated using vocal grunts and a text-to-speech augmentative and alternative communication (AAC) tablet operated with the right hand. *Eye movements:* He reported fatigue linked to his disease progression when visually fixating for prolonged times, but otherwise reported no eye movement abnormalities or visual defects.

PB1 *Diagnosis:* 41 y.o., originally left-handed male diagnosed with Friedreich’s Ataxia (FRDA) 29 years prior, at age 12. Symptoms resulted in slowed, effortful body and limb movement and affected fine motor control, and most severely affected the left side which made him effectively right handed. *Presentation:* PB1 presented in an electric wheelchair, which was positioned in front of a table with the experimental setup. *Communication:* Verbal communication was possible, but dysarthria resulted in slowed, slurred speech. He was able to effectively control a PC and a smartphone, but writing was difficult. *Eye movements:* PB1 reported impaired visual pursuit of moving targets and fatigue and discomfort fixating. Indeed, we noticed effort when making eye movements, which resulted in large muscular artifacts registered in the EEG and electrooculogram (EOG) Additionally, he reported minor nystagmus which we also noticed in the horizontal EOG.

PB2 *Diagnosis:* 43 y.o., right-handed female diagnosed with FRDA symptom-free at age 9, with symptom onset at age 12, 31 years prior to the experiment. Symptoms resulted in slowed, effortful body and limb movement and affected fine motor control. *Communication:* Dysarthric, characterized by large effort necessary to speak. Hand signaling was also often used, but also required effort. *Eye movements:* PB2 was affected by especially severe horizontal eye oscillations with onset 10 years prior to the experiment. Fixating

was also fatiguing and uncomfortable, especially when not wearing glasses or lenses.

PB4 *Diagnosis:* 48 y.o., right-handed male diagnosed with FRDA 30 years prior at age 18. *Presentation:* PB4 was not strictly wheelchair-bound. *Communication:* Speech was relatively fluent, only slightly slurred, leading to mild dysarthria. *Eye movements:* Pursuit and fixation were impaired, as characterized by self-reported fatigue and discomfort. Nystagmus. PB4 wore no corrective glasses lenses.

PC2 *Diagnosis:* 43 y.o., originally right-handed male diagnosed with Locked-in Syndrome (LiS) due to ischemic brain stem stroke 6 years prior, at age 37. He underwent a tracheostomy. *Presentation:* PC2 was seated in an electric wheelchair controlled by a caregiver, and his head was fixated in a headrest. This headrest contacted the EEG cap occipitally and centroposteriorly, which interfered slightly with cap and electrode positioning and optimization. Additionally, heavy movement artifacts were noticed in the EEG due to the combination of breathing, aspiration, the headrest and maintaining correct positioning for the experiment. The EEG cap was held in place using a chest strap. *Communication:* PC2 was completely anarthric and had no preserved facial movement. He communicated with caregivers by responding to prompted letters and words with upwards eye movement signals. Additionally, he sometimes made use of an emergency button which he could control with his left hand, the only extremity in which he had some residual control capability. *Eye movements:* Right eye: preserved vertical movement, paralysis of laterality to the right and to the left, which resulted in the inability to perform horizontal movements. Left eye: preserved vertical movement, adduction possible, abduction deficit. Deviation in the left eye resulting in diplopia, corrected by a prism glass. Self-reported tremors and fixation fatigue and discomfort. Eye tracker calibration did not succeed.

PC3 *Diagnosis:* 43 y.o., right-handed female diagnosed with hemorrhagic stroke 4 years prior, at age 39. She was tetraplegic and underwent a tracheostomy. *Presentation:* PC3 was seated in an electric wheelchair. *Communication:* PC3 was completely anarthric, but had sufficient residual motor control to indicate letters and symbols on a letterboard by pointing with movements of the right index finger and hand. *Eye movements:* Right eye: complete ptosis. When corrected with glasses, downward movement was preserved and incyclotorsion was possible through preserved cranial nerve IV. Ophthalmoplegia in other directions. Left eye: limited adduction, complete verticality, horizontal nystagmus. Intermittent nystagmus. Self-reported involuntary saccades and fixation fatigue or discomfort.

PC4 *Diagnosis:* 54 y.o., originally right-handed male diagnosed with ischemic brainstem stroke 11 years prior, at age 53. He is tetraplegic and underwent a tracheostomy. Symptoms most pronounced on left side. *Presentation:* The tracheostomy resulted in frequent aspiration which interfered with the experiment

611 and recording. This and other factors related to the condition and positioning in the wheelchair resulted
612 in large and frequent muscular and movement artifacts in the EEG. Head bobbing when concentrating on
613 the experimental task. *Communication:* PC4 had sufficient residual motor control to indicate letters and
614 symbols on a letterboard by pointing with movements of the right index finger and hand. *Eye movement:*
615 A corneal abcess in the left eye affected it's motility and resulted in deviation. The left eye had poor vision
616 (only able to distinguish shapes) and could not be closed. The right eye was not affected. The deviation
617 resulted in fatigue and discomfort fixating. Hypermetropia. The participant wore corrective glasses. Eye
618 tracker calibration did not succeed.